PHASE 1 — FOUNDATION

Soldering Bootcamp

Every kid solders something.

Say Watt Robotics · ROV Team | 2026–2027 MATE ROV Season

CLASS 1.2 · ONE SESSION · MADISON, ALABAMA

THE MEETING

Three hours, all hands on irons.

0:00	**Welcome & Safety Brief**	15 min
0:15	**Why Soldering Matters**	15 min
0:30	**Recap & the five moves**	25 min
0:55	**Hands-on practice joints**	60 min
1:55	*Break*	10 min
2:05	**Practice — every kid steady**	30 min
2:35	**Inspect & certify joints**	15 min
2:50	**Wrap-up & homework**	10 min

TODAY'S ROLES

Four jobs. Four students. Different every week.

Every meeting we assign these four — by season's end, every student has done every job.

NOTETAKER / SCRIBE

TODAY: ________________________________

Captures every decision, action item, and blocker. Files notes with the CFO for the engineering notebook.

SAFETY BRIEFER

TODAY: ________________________________

Delivers the standing four-question safety brief at the start of the meeting.

PHOTOGRAPHER

TODAY: ________________________________

Captures photos and short videos throughout — feeds the engineering notebook and the marketing display.

TIMEKEEPER

TODAY: ________________________________

Holds the meeting to the agenda's clock. Calls the time at each block change.

01

SECTION 01

Welcome & Safety Brief

0:00 – 0:15 · 15 MIN

Every meeting we ever hold will begin in exactly this place.

THE STANDING RITUAL

Every meeting opens with a safety brief.

Why we do this

Real engineering and dive teams brief safety before every job — not because something went wrong, but so it never does. We adopt the same habit on day one. Sixty seconds, every meeting, no exceptions.

Each brief answers the same four questions, so the team always knows what to expect.

1. What are we doing today?
2. What could hurt someone?
3. How do we prevent it?
4. Who do we tell if something happens?

TODAY'S BRIEF · CLASS 1.2

THE WORK

First solder joints, the kit unboxing, and the first real electronics on the robot.

THE HAZARDS

Soldering irons at roughly 700°F. Burns, lead and flux fumes, and hot solder spatter.

PREVENTION

Irons in their stands, never on the bench. Fume fan on. Safety glasses down, hair tied back. Wash hands after handling solder.

IF SOMETHING HAPPENS

Cool a burn under running water, tell the advisor, and log it. Every burn gets reported — even a small one.

LOOKING AHEAD

Class 1.3 — the kit comes out of the box and takes its first electronics.

SECTION 02

Why We Solder

0:15 – 0:30 · 15 MIN

02

Before anyone touches the kit, everyone understands why a good joint matters.

WHY WE SOLDER

A robot is only as good as its weakest joint.

Soldering fuses two wires into one connection that holds — electrically and mechanically. Every thruster, sensor, and camera on our ROV reaches the control box through a joint someone made by hand.

A good joint is shiny, solid, and stronger than the wire around it. A bad one — cold, dull, or barely touching — works on the bench, then fails in the water, mid-mission, where we cannot reach it.

This is the first real engineering skill of the season. Today, every kid earns it.

THE STANDARD

Shiny, solid, and built to last the season.

A joint you would trust at 2.5 m

In the dark, with the mission clock running, where nobody can fix it.

KNOW YOUR JOINT

Three joints. Know them on sight.

COLD JOINT

Dull and grainy

Too little heat. The solder sits on top in a dull blob instead of flowing in. Looks done, fails later. Reflow it.

GOOD JOINT

Shiny and smooth

Heat on the joint, solder wicks around the wire in a bright cone. Strong and clean. This is the target.

BRIDGED JOINT

Touching its neighbor

Too much solder jumps to the next pad and shorts them together. Wick the extra away and redo it.

Every kid makes all three today — and learns to tell them apart by eye.

SECTION 03

Soldering Bootcamp

0:30 – 1:55 · 85 MIN

Watch it done, see it done, then do it — every kid, hands on an iron.

03

BEFORE CLASS

Watch it at home. Build it in class.

The concepts come from MATE’s own instructors. Watch these short presentations before class, so every minute of our meeting is spent at the bench, not at a screen.

As you watch, notice: how to tin the tip; why the heat goes on the joint, not the solder; what a good joint looks like; and how heat-shrink and strain relief protect the wire.

Come ready to solder — in class we recap, demo once, and then it’s hands on irons.

PRE-WATCH HOMEWORK

Three short videos, then straight to the iron.

MATE: Soldering & Waterproofing Wires

Plus MATE: Using Tools (20–30 min) and How to Solder (22 min) — all assigned on the team site.

THE DEMONSTRATION

The moves, start to finish.

1 Strip

BARE THE WIRE

Strip about 5 mm of insulation. Clean, bright copper — no nicks in the strands.

2 Tin the tip

PREP THE IRON

Wet sponge, then a fresh dab of solder on the tip. Do this before every single joint.

3 Heat the joint

IRON ON METAL

Press the iron to the wires, not the solder. Count two seconds to let them heat.

4 Feed the solder

LET IT FLOW

Touch solder to the hot joint, not the iron. It melts and wicks in. Then stop.

5 Heat-shrink

SEAL THE JOINT

Slide shrink tube over the joint and heat it. Add a service loop for strain relief.

6 Inspect

CHECK YOUR WORK

Hold it to the light. Shiny and solid? Done. Dull or lumpy? Reflow and try again.

HANDS ON

Now everyone makes one.

1. **Strip and tin a wire**
   Each kid strips a wire and tins it. We check for clean copper and a shiny tip before moving on.
2. **Make your first joint**
   Join two tinned wires. Heat the joint, feed the solder, then let it cool without moving it.
3. **Heat-shrink and strain relief**
   Slide the tube, shrink it, and add a service loop so the wire never pulls on the joint itself.
4. **Inspect and redo**
   Hold it to the light. Shiny and solid moves on; dull or lumpy gets reflowed. Redoing it is the lesson.

Two things to remember. No one opens the kit until they have made one joint they are proud of. The redo is not failure — it is the skill arriving.

HOW WE LEARN

Teach once. Then step back.

> ***“We show a skill one time — then we step back and let them struggle to it. The struggle is where the engineer is made.”***

MATE’s Section 4.3 is strict: the work has to be the students’. Adults teach, demonstrate, and keep the room safe — but adults do not build the robot, write the documentation, or hold the iron through the joint.

Parents demonstrate

Show the move once, then photograph and encourage. Hands stay off the work — even when it is slow.

Students do the work

Every joint, every fix, every redo is theirs. Slow and real beats fast and borrowed.

Help is disclosed

Any outside help — including AI — gets written into the documentation. Honesty is scored in April.

BREAK

Ten minutes.

We resume at 1:55 — hands washed, irons cooling, and we start on time.

SECTION 04

Wrap-Up & Homework

2:50 – 3:00 · 10 MIN

04

Close the loop: what each student carries out, and what the team carries forward into the water.

WRAP-UP & HOMEWORK

Before we meet again.

EVERY STUDENT

1. Practice one clean joint at home — shiny, solid, strain-relieved.
2. Be able to spot a cold joint vs. a good one on sight.
3. Closed-toe shoes again next time — the kit comes out.

THE PARALLEL TRACK · LEADS & ADVISORS

Battery, charger, and leads in hand before the electronics class. SeaMATE lab kits stocked. Pool prerequisites in motion — lifeguard, swim checks, host rules in writing.

NEXT LESSON · CLASS 1.3

Kit Unboxing & Electronics

The Triggerfish comes out of the box, gets inventoried, and takes its first real connections — plus the team's records and capsule designs open.

ACTION ITEMS

What we decided. Who owns it. When it's due.

The Notetaker fills these in during the meeting and reads them back before we leave.

ITEM	OWNER	DUE

BEFORE WE LEAVE

Tools accounted for. Shop reset.

We do this together, every meeting. No exceptions — the Tool Steward leads.

- [ ] Soldering irons unplugged, cool, and put away.
- [ ] All hand tools and test equipment back in their bins.
- [ ] Batteries off charge; charger stations powered down.
- [ ] Workshop swept; benches wiped; trash out.
- [ ] Today's photos uploaded to the team folder.
- [ ] Notetaker's meeting notes filed with the CFO for the engineering notebook.

SIX KIDS, SIX IRONS

Now it’s real.

Everyone soldered. The kit is open, the first joints are on the robot, and the capsule is on paper. Next time, the kit comes out.